

Quantum-Assisted Multi-Asset Portfolio Construction

Step 02

Quantum-Compatible QUBO and Ising Formulation

From continuous portfolio economics to a fixed-cardinality QAOA active-set problem

Contents

1	Why a quantum-compatible reformulation is needed	3
2	Binary variables and the trade map	3
3	Deriving the QUBO from a quadratic economic core	3
4	Term-by-term financial derivation	4
4.1	Expected growth and income	4
4.2	Covariance risk	4
4.3	Concentration and market impact	4
4.4	Linear trading cost	4
5	Scenario-warning hinge: exact cases and surrogate case	4
6	The implemented active-selection proxy	5
7	Fixed cardinality	5
8	QUBO matrix convention	5
9	Mapping to an Ising Hamiltonian	6
10	Feasible initial state and XY mixer	6
11	The QAOA state and optimization loop	7
12	Decode, reoptimize, and audit	7
13	Observed result	7
14	Worked three-bit example	7
15	Reproduction checklist and claim boundary	8
A	From continuous trades to a binary active sleeve	8
B	General quadratic substitution	8
C	Deriving important QUBO components	9
C.1	Expected return and income	9

C.2 Covariance risk	9
C.3 Concentration and impact	9
C.4 Linear transaction cost	9
D Scenario-warning hinge in binary space	9
E Fixed-cardinality constraint	10
F QUBO conventions and the factor-of-two test	10
G Full Ising mapping	10
H QAOA state preparation and operators	11
I Exact enumeration and scale	11
J Worked four-bit QUBO example	12
K Exact post-processing contract	12
L Reproduction tests	12

1 Why a quantum-compatible reformulation is needed

Step 1 uses continuous weights. Gate-based QAOA is naturally designed for binary decisions. The project therefore does not attempt to encode all 50 real-valued weights directly into qubits. Instead it asks a narrower question:

Which candidate assets should be allowed to change while a classical optimizer determines the exact trade sizes?

This is an active-rebalancing design. Assets outside the selected sleeve remain at their incumbent weights. A selected asset receives permission to trade; it is not forced to receive a nonzero trade.

2 Binary variables and the trade map

Let the reduced universe contain m candidate trade coordinates. Define

$$z_j \in \{0, 1\}, \quad j = 1, \dots, m, \quad (1)$$

where $z_j = 1$ selects candidate j . In a general trade-map derivation, let $d_j \in \mathbb{R}^N$ be the proposed signed change associated with candidate j , and form

$$D = [d_1 \ d_2 \ \dots \ d_m] \in \mathbb{R}^{N \times m}. \quad (2)$$

Starting from a feasible reference portfolio $\bar{w}$,

$$w(z) = \bar{w} + Dz. \quad (3)$$

If every candidate trade is self-financing, $\mathbf{1}^\top d_j = 0$, then

$$\mathbf{1}^\top w(z) = \mathbf{1}^\top \bar{w} + \mathbf{1}^\top Dz = 1. \quad (4)$$

Thus budget feasibility can be built into the candidate design.

3 Deriving the QUBO from a quadratic economic core

Suppose the continuous objective can be written as

$$F(w) = w^\top H w + r^\top w + c, \quad (5)$$

where $H = H^\top \succeq 0$. Substitute $w = \bar{w} + Dz$:

$$F(\bar{w} + Dz) = (\bar{w} + Dz)^\top H (\bar{w} + Dz) + r^\top (\bar{w} + Dz) + c \quad (6)$$

$$= \bar{w}^\top H \bar{w} + 2z^\top D^\top H \bar{w} + z^\top D^\top H Dz \quad (7)$$

$$+ r^\top \bar{w} + r^\top Dz + c. \quad (8)$$

Collecting constant, linear, and quadratic terms gives

$$F(z) = c_Q + q^\top z + z^\top Q z, \quad (9)$$

with

$$Q = D^\top H D, \quad (10)$$

$$q = 2D^\top H \bar{w} + D^\top r, \quad (11)$$

$$c_Q = \bar{w}^\top H \bar{w} + r^\top \bar{w} + c. \quad (12)$$

This is the central algebraic bridge from the portfolio model to a QUBO.

4 Term-by-term financial derivation

4.1 Expected growth and income

A linear reward $-a^\top w$ becomes

$$-a^\top(\bar{w} + Dz) = -a^\top\bar{w} - (D^\top a)^\top z. \quad (13)$$

It contributes only a linear binary coefficient.

4.2 Covariance risk

The variance penalty is

$$(\bar{w} + Dz)^\top \Sigma (\bar{w} + Dz) = \bar{w}^\top \Sigma \bar{w} + 2z^\top D^\top \Sigma \bar{w} + z^\top D^\top \Sigma Dz. \quad (14)$$

The last term creates pairwise binary interactions. Two individually attractive trades may be unattractive together if their joint covariance contribution is large.

4.3 Concentration and market impact

The same expansion applies to $w^\top w$ and $(w - w^0)^\top \Lambda (w - w^0)$. Therefore, concentration and quadratic impact naturally generate QUBO couplings.

4.4 Linear trading cost

If candidate directions are fixed, the absolute trade associated with candidate j is known in advance. Its linear cost can be stored as κ_j , giving

$$C_{\text{lin}}(z) = C_0 + \sum_j \kappa_j z_j. \quad (15)$$

In the implemented active-screening proxy, the cost of selecting versus not selecting a coordinate is calculated relative to the incumbent.

5 Scenario-warning hinge: exact cases and surrogate case

For scenario s ,

$$L_s(w(z)) - \tau_s = b_s + v_s^\top z, \quad (16)$$

where $b_s = \ell_s^\top \bar{w} - \tau_s$ and $v_s = D^\top \ell_s$. The exact penalty is

$$\omega_s \left[\max\{b_s + v_s^\top z, 0\} \right]^2. \quad (17)$$

There are three cases:

1. **Always inactive:** $b_s + v_s^\top z \leq 0$ for every feasible z ; the penalty is zero.
2. **Always active:** $b_s + v_s^\top z \geq 0$ for every feasible z ; the square expands exactly into a QUBO.

3. **Crossing:** some supports are below and some above the warning. A single quadratic polynomial cannot reproduce the kink exactly without auxiliary variables. The project may use a local quadratic proxy for candidate generation, but the exact hinge is recomputed after decoding.

This is why the binary energy is a search model, not the final financial certificate.

6 The executed active-rebalance QUBO

The final evidence path uses $m = 11$ screened coordinates and $k = 6$. A bit is permission for a coordinate to move in the later continuous solve; it is not a fixed executed trade vector. The active QUBO is therefore a marginal-utility support-ranking surrogate rather than the full Step 1 objective.

Let s_i be the screened active-selection score, t_i the signed proxy trade, and c_i the normalized implementation-cost burden. After normalization, the executed coefficients are

$$q_i = -\tilde{s}_i^2 + 0.05 \tilde{c}_i \tilde{s}_i, \quad (18)$$

$$Q = \lambda_b \tilde{t} \tilde{t}^\top + 0.04 (\tilde{t} \tilde{t}^\top) \odot \tilde{\Sigma} + 0.02 (\tilde{t} \tilde{t}^\top) \odot \tilde{\Gamma}, \quad (19)$$

$$E_{\text{active}}(z) = q^\top z + z^\top Q z. \quad (20)$$

The final run uses $\lambda_b = 1.0$. Scenario co-loss, concentration, class balance, exact absolute-value cost, and the full policy rulebook are not separate terms in this active QUBO; they remain in screening and/or exact downstream refinement.

7 Fixed cardinality

The final experiment imposes

$$\sum_{j=1}^{11} z_j = 6. \quad (21)$$

The number of feasible supports is

$$\binom{11}{6} = \frac{11!}{6!5!} = 462. \quad (22)$$

A penalty-only encoding could add

$$P \left(\sum_j z_j - k \right)^2, \quad (23)$$

which expands to linear and pairwise terms. The implemented Qiskit program also includes the exact equality constraint and uses a Hamming-weight-preserving circuit, reducing the need to search invalid cardinalities.

8 QUBO matrix convention

Because $z_i^2 = z_i$ for binary variables, a symmetric-matrix representation

$$E(z) = z^\top Q z + q^\top z + c \quad (24)$$

corresponds to the upper-triangular polynomial

$$E(z) = c + \sum_i (Q_{ii} + q_i) z_i + \sum_{i < j} 2Q_{ij} z_i z_j. \quad (25)$$

The factor of two on off-diagonal terms is a common source of implementation errors. The project explicitly validates the matrix and polynomial energies on bitstrings.

9 Mapping to an Ising Hamiltonian

Write the polynomial as

$$E(z) = c + \sum_i a_i z_i + \sum_{i < j} b_{ij} z_i z_j. \quad (26)$$

Use the spin mapping

$$z_i = \frac{1 - s_i}{2}, \quad s_i \in \{-1, +1\}. \quad (27)$$

Then

$$z_i z_j = \frac{1 - s_i - s_j + s_i s_j}{4}. \quad (28)$$

After collecting terms,

$$E(s) = C + \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j, \quad (29)$$

where

$$J_{ij} = \frac{b_{ij}}{4}, \quad (30)$$

$$h_i = -\frac{a_i}{2} - \frac{1}{4} \sum_{j \neq i} b_{ij}, \quad (31)$$

$$C = c + \frac{1}{2} \sum_i a_i + \frac{1}{4} \sum_{i < j} b_{ij}. \quad (32)$$

Replacing s_i by Pauli Z_i gives the cost Hamiltonian

$$H_C = CI + \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j. \quad (33)$$

The constant does not change the minimizing bitstring, but it must be retained when checking numerical energy equality.

10 Feasible initial state and XY mixer

The initial state is the equal-amplitude Dicke state over all six-of-eleven bitstrings:

$$|D_{11}^6\rangle = \frac{1}{\sqrt{\binom{11}{6}}} \sum_{|z|=6} |z\rangle. \quad (34)$$

The ring mixer is

$$H_M = \frac{1}{2} \sum_{i=1}^{11} (X_i X_{i+1} + Y_i Y_{i+1}), \quad (35)$$

with periodic indexing. It exchanges excitations and preserves total Hamming weight in the noiseless model.

11 The QAOA state and optimization loop

At depth p , the QAOA state is

$$|\psi(\gamma, \beta)\rangle = \prod_{\ell=1}^p e^{-i\beta_{\ell}H_M} e^{-i\gamma_{\ell}H_C} |D_{11}^6\rangle. \quad (36)$$

The final experiment used $p = 2$, 1,024 shots, 60 classical optimizer iterations, and 20 independent seeds. A classical optimizer updates (γ, β) using sampled energy information.

12 Decode, reoptimize, and audit

For every promising measured bitstring:

1. Decode the six selected candidate tickers.
2. Hold all inactive assets at incumbent weights.
3. Solve the exact continuous Step 5 problem over the active coordinates.
4. Recompute return, covariance risk, transaction cost, scenario losses, and all hard constraints.
5. Reject any hard-breaching support.
6. Compare with exact enumeration, greedy selection, and local search.

13 Observed result

The selected support was

$$\{\text{SGOV}, \text{BIL}, \text{SHY}, \text{QQQ}, \text{VUG}, \text{MTUM}\}. \quad (37)$$

The continuous optimizer executed four material trades: buy SGOV and sell MTUM, QQQ, and VUG. BIL and SHY were permissions that optimally received no material change.

The best selected QAOA energy matched the exact minimum, with zero energy gap. Fifteen of twenty seeds reached the exact reduced optimum, a 75% exact-hit rate. Because all 462 states are classically enumerable, this establishes correctness of the selected support, not quantum advantage.

14 Worked three-bit example

For three candidates with exactly two selected, the feasible bitstrings are 110, 101, 011. If

$$E(z) = -2z_1 - z_2 - 1.5z_3 + 0.8z_1z_2 + 0.1z_1z_3 + 0.5z_2z_3, \quad (38)$$

Direct evaluation gives one energy per feasible support. At project scale the same calculation enumerates all 462 reduced-QUBO energies. Only retained QAOA and high-ranked supports are then passed to the 50-asset continuous refinement; the enumeration itself does not continuously reoptimize all 462 supports.

15 Reproduction checklist and claim boundary

Reproduction requires the ordered 11-ticker universe, the complete Q, q, c convention, cardinality 6, circuit depth, shot count, optimizer settings, seed list, measured distributions, exact enumeration, and continuous refinements. The defensible claim is: the selected best-of-20 QAOA run found the exact optimum of the reduced 6-of-11 QUBO and produced a hard-compliant refined portfolio. The project does not claim speedup, hardware advantage, or superiority to classical optimization.

A From continuous trades to a binary active sleeve

The full portfolio has 50 continuous weights. The implemented quantum subproblem does not discretize all 50 weights. Instead, it begins with an incumbent $\bar{w}$ and constructs $M = 11$ candidate trade directions. Let $d_j \in \mathbb{R}^{50}$ be the weight change associated with enabling candidate j , and collect the directions in

$$D = \begin{bmatrix} d_1 & d_2 & \cdots & d_M \end{bmatrix}. \quad (39)$$

A bit $z_j \in \{0, 1\}$ records whether candidate j is selected. The approximate portfolio represented by a bitstring is

$$w(z) = \bar{w} + Dz. \quad (40)$$

Each column is constructed or screened so that the trade has a clear financial interpretation. Budget-neutral directions satisfy

$$\mathbf{1}^\top d_j = 0, \quad (41)$$

and therefore every bitstring preserves the budget:

$$\mathbf{1}^\top w(z) = \mathbf{1}^\top \bar{w} + \mathbf{1}^\top Dz = 1. \quad (42)$$

In the implemented active-rebalancing design, the binary support is ultimately a permission set rather than a fixed trade vector. After selection, the exact continuous optimizer chooses the trade sizes.

B General quadratic substitution

Suppose the quadratic part of the Step 1 objective is

$$f(w) = w^\top Hw + r^\top w + c, \quad (43)$$

where $H \succeq 0$. Substitute $w = \bar{w} + Dz$:

$$f(\bar{w} + Dz) = (\bar{w} + Dz)^\top H(\bar{w} + Dz) + r^\top (\bar{w} + Dz) + c \quad (44)$$

$$= z^\top D^\top H Dz + \left(2D^\top H \bar{w} + D^\top r \right)^\top z + \bar{w}^\top H \bar{w} + r^\top \bar{w} + c. \quad (45)$$

The constant can be ignored when ranking bitstrings. Therefore

$$Q_{\text{core}} = D^\top H D, \quad q_{\text{core}} = 2D^\top H \bar{w} + D^\top r. \quad (46)$$

This identity is the mathematical bridge from the continuous economic model to a QUBO.

C Deriving important QUBO components

C.1 Expected return and income

A linear reward $-a^\top w$ becomes

$$-a^\top(\bar{w} + Dz) = \text{constant} - (D^\top a)^\top z. \quad (47)$$

Thus an attractive candidate receives a negative linear energy contribution.

C.2 Covariance risk

Variance becomes

$$(\bar{w} + Dz)^\top \Sigma (\bar{w} + Dz) = z^\top D^\top \Sigma Dz + 2\bar{w}^\top \Sigma Dz + \text{constant}. \quad (48)$$

The diagonal entries measure each candidate's incremental risk. The off-diagonal entries measure whether two candidates reinforce or diversify one another.

C.3 Concentration and impact

Concentration uses $H = I$. Quadratic impact uses $H = \Lambda$. Both therefore have the same substitution pattern. This is useful because a single tested matrix-expansion routine can generate several economic components.

C.4 Linear transaction cost

A pure linear cost in a predeclared candidate trade size can be inserted in q . In the final project, exact execution cost is recomputed after continuous refinement. The QUBO cost term is therefore a ranking signal, not the final accounting authority.

D Scenario-warning hinge in binary space

For scenario vector ℓ_s , define

$$b_s = \ell_s^\top \bar{w} - \tau_s, \quad v_s = D^\top \ell_s. \quad (49)$$

The warning excess is

$$h_s(z) = \max\{b_s + v_s^\top z, 0\}. \quad (50)$$

The squared penalty is piecewise quadratic. There are three cases:

1. If $b_s + v_s^\top z \leq 0$ for every feasible bitstring, the penalty is always zero.
2. If $b_s + v_s^\top z \geq 0$ for every feasible bitstring, the max may be removed and the penalty expands exactly into a QUBO.
3. If the expression changes sign across feasible bitstrings, one quadratic polynomial cannot exactly represent the hinge without extra variables. The project may use a local quadratic surrogate for candidate generation, but recomputes the exact hinge before accepting a portfolio.

This distinction prevents a surrogate from being mistaken for a financial feasibility certificate.

E Fixed-cardinality constraint

The experiment selects exactly $K = 6$ of $M = 11$ candidates:

$$\sum_{j=1}^M z_j = K. \quad (51)$$

A conventional penalty is

$$P \left(\sum_j z_j - K \right)^2. \quad (52)$$

Using $z_j^2 = z_j$ for binary variables,

$$P \left(\sum_j z_j - K \right)^2 = P \left[\sum_j z_j + 2 \sum_{i < j} z_i z_j - 2K \sum_j z_j + K^2 \right] \quad (53)$$

$$= P(1 - 2K) \sum_j z_j + 2P \sum_{i < j} z_i z_j + PK^2. \quad (54)$$

The constant can be dropped. In the final circuit, a fixed-Hamming-weight initial state and an XY mixer preserve cardinality, so the search remains in the feasible subspace and the penalty is not the only enforcement mechanism.

F QUBO conventions and the factor-of-two test

Two equivalent conventions are common:

$$E(z) = z^T Q z + q^T z, \quad (55)$$

$$E(z) = \sum_i a_i z_i + \sum_{i < j} b_{ij} z_i z_j. \quad (56)$$

For a symmetric matrix, $z^T Q z$ contains $2Q_{ij}z_i z_j$ for $i < j$. Therefore the conversion requires

$$Q_{ij} = \frac{1}{2} b_{ij} \quad (i \neq j). \quad (57)$$

A mandatory unit test enumerates small bitstrings and verifies that both conventions return identical energy.

G Full Ising mapping

Use the binary-to-spin substitution

$$z_i = \frac{1 - s_i}{2}, \quad s_i \in \{-1, +1\}. \quad (58)$$

In a quantum Hamiltonian, s_i becomes the eigenvalue of Z_i . For the upper-triangular polynomial

$$E(z) = c + \sum_i a_i z_i + \sum_{i < j} b_{ij} z_i z_j, \quad (59)$$

substitution gives

$$H_C = \kappa I + \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j, \quad (60)$$

with

$$J_{ij} = \frac{1}{4} b_{ij}, \quad (61)$$

$$h_i = -\frac{1}{2} a_i - \frac{1}{4} \sum_{j \neq i} b_{ij}, \quad (62)$$

$$\kappa = c + \frac{1}{2} \sum_i a_i + \frac{1}{4} \sum_{i < j} b_{ij}. \quad (63)$$

The offset κ does not change the minimizing bitstring but must be retained when comparing QUBO energies with Hamiltonian eigenvalues.

H QAOA state preparation and operators

A feasible initial state is a Dicke state over all bitstrings with Hamming weight K :

$$|D_K^M\rangle = \binom{M}{K}^{-1/2} \sum_{|z|=K} |z\rangle. \quad (64)$$

The cost unitary is

$$U_C(\gamma) = e^{-i\gamma H_C}. \quad (65)$$

A number-preserving XY mixer can be written on a ring as

$$H_M = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} (X_i X_j + Y_i Y_j), \quad U_M(\beta) = e^{-i\beta H_M}. \quad (66)$$

Because H_M commutes with total excitation number, it moves probability among feasible six-of-eleven states without leaving the cardinality sector.

At depth p , the state is

$$|\psi(\gamma, \beta)\rangle = \prod_{r=1}^p U_M(\beta_r) U_C(\gamma_r) |D_K^M\rangle. \quad (67)$$

A classical optimizer updates the angles to reduce the sampled expectation of H_C .

I Exact enumeration and scale

The number of feasible supports is

$$\binom{11}{6} = 462. \quad (68)$$

Every feasible six-of-eleven state can therefore be evaluated classically for the executed reduced-QUBO energy. Exact enumeration establishes the ground state and measures QAOA seed reliability. The best-of-20 run reached that exact QUBO minimum, and 15 of 20 seeds did so. This does not prove exhaustive optimality of the continuously refined financial objective or computational advantage.

J Worked four-bit QUBO example

Suppose four candidates must select two. Let stand-alone coefficients be $a = (-3, -2, -1, -1)$ and let one undesirable pair have $b_{12} = 4$, while other pair terms are zero. Ignoring cardinality, candidates 1 and 2 look best individually, but together their energy is

$$E(1100) = -3 - 2 + 4 = -1. \quad (69)$$

Candidates 1 and 3 have

$$E(1010) = -3 - 1 = -4. \quad (70)$$

The pairwise term changes the preferred bundle. This is the essential reason a combinatorial optimizer can be useful: assets and trades cannot always be ranked independently.

K Exact post-processing contract

For every measured bitstring, the implementation must:

1. confirm Hamming weight six;
2. decode the selected tickers;
3. rebuild the exact active-support continuous problem;
4. solve for continuous weights and trades;
5. recompute return, covariance risk, cost, impact, and exact scenario losses;
7. compare the support against exact reduced-QUBO enumeration and classical heuristics; and
8. report the portfolio only if the full financial model passes.

The QUBO is therefore a candidate generator. The continuous model remains the source of truth.

L Reproduction tests

A faithful reproduction should test budget preservation for each direction, QUBO expansion against the continuous quadratic core, symmetric-versus-polynomial conventions, QUBO-to-Ising energy equality, cardinality preservation, exact reduced-QUBO enumeration rank, decoding order, full-constraint reoptimization, and seed-level reliability. Random seeds, depth, shots, optimizer iterations, backend, initial state, mixer, and Hamiltonian scaling must be stored.